

Hierarchical models Exercises

Exercise 1 Modeling the Impact of Climate Change

Using the **ITEX diversity dataset**, think about a research question you could ask regarding the impact of **one aspect of climate change** on a **variable of interest**, across different sites.

Explore your research question using modeling tools you learned in today's lecture.

Your Tasks:

1. Model Planning:

- What is your **response variable**?
- Which variables make sense as **fixed effects**?
- Which variable(s) should be treated as **random effects** (e.g., site, year)?

2. Fit the Model:

- Use `lmer()` to fit a model to the data and look at the output with `summary()`.

3. Reflect:

- Does the model structure reflect **how the data was collected**?

Exercise 2 (optional)

What is the impact of climate change on your variable of interest **within sites**?

Now that you've explored overall patterns, focus your model on how **climate change impacts your variable of interest, within each site**.

Your Tasks:

1.
 - What changes when we ask about relationships *within sites* instead of *across* them?
 - Should you adjust the way **site** is treated in your model?
2. Model Check:

3. **Fit a New Model (if needed):**

- Update your `lmer()` call to reflect this new focus.
- Reinterpret the output